

Chapter 22

Maximum Likelihood Estimation

Lecture 22 | Devore 9e: 6.2

The method of moments matched summary statistics. Maximum likelihood asks a different and more direct question:

Which parameter value makes the data I actually saw as likely as possible?

22.1 A discrete warm-up

Example 22.1 (Candy jar). A jar holds 10 candies, k of them red, with k unknown. You draw 4 candies without replacement and observe 3 red. Estimate k .

A first instinct is to match proportions: $\frac{k}{10} = \frac{3}{4}$, so $k = 7.5$ — which is not an integer, and not obviously right. Instead, write down the probability of what happened, as a function of k :

$$\mathbb{P}(3 \text{ red out of } 4) = \frac{\binom{k}{3} \binom{10-k}{1}}{\binom{10}{4}}, \quad 3 \leq k \leq 9.$$

(The range is forced: we saw 3 reds so $k \geq 3$, and we saw a non-red so $k \leq 9$.) The denominator does not involve k , and $\binom{k}{3} \binom{10-k}{1} = \frac{k(k-1)(k-2)}{6} (10-k)$, so we maximize $k(k-1)(k-2)(10-k)$ over the integers:

k	3	4	5	6	7	8	9
$k(k-1)(k-2)(10-k)$	42	144	300	480	630	672	504

So $\hat{k}_{\text{MLE}} = 8$.

Remark 22.2. The estimate is the value of the parameter under which the observed data is least surprising. Note that this required no derivative — for a discrete parameter you simply tabulate.

22.2 The continuous case, and the likelihood function

Example 22.3 (Light bulbs). Lifetimes follow $\text{Exponential}(\lambda)$. You observe 5 bulbs:

$$x_1 = 2, \quad x_2 = 1.5, \quad x_3 = 3, \quad x_4 = 0.5, \quad x_5 = 4.$$

Estimate λ .

We cannot copy the discrete recipe, because $\mathbb{P}\{X_1 = 2, X_2 = 1.5, \dots\} = 0$ for a continuous variable. Use the joint *density* instead: it plays exactly the role that probability played above.

Definition 22.4 (Likelihood function). Let $X_1, \dots, X_n$ be a sample from a distribution with parameters $\theta_1, \dots, \theta_m$, and let $x_1, \dots, x_n$ be the observed data. The **likelihood function** is the joint pdf (or pmf) viewed as a function of the parameters:

$$L(\theta_1, \dots, \theta_m) := f(x_1, x_2, \dots, x_n; \theta_1, \dots, \theta_m).$$

The **maximum likelihood estimates** $\hat{\theta}_1, \dots, \hat{\theta}_m$ are the values maximizing L .

Remark 22.5. Read the roles carefully. The joint pdf is normally a function of the data with the parameters fixed. The likelihood is the *same formula* regarded as a function of the parameters with the data fixed. Same expression, opposite point of view.

22.2.1 The log trick

For an i.i.d. sample the joint density is a product, $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$, and products are unpleasant to differentiate. Since log is strictly increasing, maximizing L is the same as maximizing

$$\ell(\theta) := \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta),$$

which turns the product into a sum. Always take logs first.

22.2.2 Exponential, worked

$$L(\lambda) = \prod_{i=1}^5 \lambda e^{-\lambda x_i}, \quad \log L(\lambda) = 5 \log \lambda - \lambda \sum_{i=1}^5 x_i.$$

Differentiate and set to zero:

$$\frac{d}{d\lambda} \log L(\lambda) = \frac{5}{\lambda} - \sum_{i=1}^5 x_i \stackrel{\text{set}}{=} 0 \quad \implies \quad \hat{\lambda}_{\text{MLE}} = \frac{5}{\sum_{i=1}^5 x_i}.$$

With $\sum x_i = 11$, $\hat{\lambda} = \frac{5}{11} = \frac{1}{2.2}$. In general

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}$$

which is reassuring, since $\mathbb{E}[X] = 1/\lambda$ for the exponential.

22.3 Normal

Example 22.6 (Unknown mean). X_i i.i.d. $\text{Normal}(\mu, 1)$ with data 8, 9, 10, 11, 12.

$$L(\mu) = \prod_{i=1}^5 \frac{1}{\sqrt{2\pi}} e^{-(x_i - \mu)^2/2}, \quad \log L(\mu) = -\frac{5}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^5 (x_i - \mu)^2.$$

Then

$$\frac{d}{d\mu} \log L(\mu) = \sum_{i=1}^5 (x_i - \mu) \stackrel{\text{set}}{=} 0 \quad \implies \quad \hat{\mu}_{\text{MLE}} = \frac{\sum_i x_i}{5} = \bar{x} = 10.$$

Remark 22.7. Notice that maximizing the likelihood here is the same as *minimizing* $\sum_i (x_i - \mu)^2$. Least squares is maximum likelihood under normal errors — that identity is the foundation of regression.

Example 22.8 (Unknown variance). X_i i.i.d. $\text{Normal}(10, \sigma^2)$ with the same data. Writing $v = \sigma^2$,

$$\log L(v) = -\frac{5}{2} \log(2\pi v) - \frac{1}{2v} \sum_{i=1}^5 (x_i - 10)^2,$$

$$\frac{d}{dv} \log L(v) = -\frac{5}{2} \cdot \frac{1}{v} + \frac{\sum_i (x_i - 10)^2}{2v^2} \stackrel{\text{set}}{=} 0 \quad \implies \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{\sum_{i=1}^5 (x_i - 10)^2}{5} = \frac{10}{5} = 2.$$

Remark 22.9 (MLE can be biased). In general, when μ is also estimated from the data, the MLE of the variance is

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2,$$

the *biased* sample variance from Lecture 20. So maximum likelihood does not automatically produce unbiased estimators. It has other virtues — consistency, asymptotic efficiency, and invariance — which is why it is the default method anyway.

22.4 When calculus does not apply

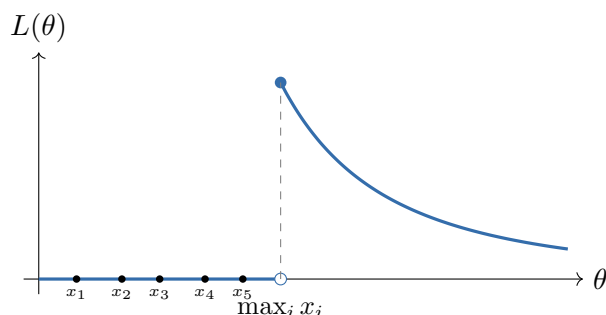
Example 22.10 (Uniform $[0, \theta]$, one more time). $X_1, \dots, X_n$ i.i.d. $\text{Uniform}[0, \theta]$. The likelihood is

$$L(\theta) = f(x_1) \cdots f(x_n) = \begin{cases} \frac{1}{\theta^n}, & 0 \leq x_i \leq \theta \text{ for all } i, \\ 0, & \text{otherwise.} \end{cases}$$

The condition “ $x_i \leq \theta$ for all i ” is just $\theta \geq \max_i x_i$. So

$$L(\theta) = \begin{cases} 0, & \theta < \max_i x_i, \\ \theta^{-n}, & \theta \geq \max_i x_i, \end{cases}$$

which is 0, jumps up at $\theta = \max_i x_i$, and then decreases.



There is no interior critical point — the derivative is never zero. The maximum sits at the left edge of the allowed region:

$$\hat{\theta}_{\text{MLE}} = \max(X_1, \dots, X_n)$$

Remark 22.11. So maximum likelihood recovers the biased estimator from Lecture 20, not the bias-corrected $\frac{n+1}{n}$ max. Two lessons:

1. Always check whether the support of the distribution depends on the parameter. If it does, do not reach for the derivative — sketch L instead.
2. MLE and unbiasedness are different criteria and can disagree.

Example 22.12 (Three heads in three flips). A coin with unknown heads probability p is flipped n times and shows k heads. The likelihood is $L(p) = p^k(1-p)^{n-k}$, and for $0 < k < n$ the log trick gives

$$\frac{d}{dp} \log L(p) = \frac{k}{p} - \frac{n-k}{1-p} \stackrel{\text{set}}{=} 0 \quad \implies \quad \hat{p}_{\text{MLE}} = \frac{k}{n}.$$

But if all three of three flips come up heads, $L(p) = p^3$ is increasing on $[0, 1]$, the derivative never vanishes, and the maximum sits at the edge: $\hat{p} = 1$.

This is a correct use of the method and a poor estimate: after three flips it declares that the coin can never land tails. A common remedy is to add one imaginary head and one imaginary tail before estimating,

$$\hat{p} = \frac{k+1}{n+2} = \frac{4}{5} \quad \text{here,}$$

which is Laplace's *rule of succession*. It is what a continuous version of Bayes' theorem (Lecture 7) gives when p is itself treated as random, uniform on $[0, 1]$: after the data, p has the Beta($k+1, n-k+1$) distribution (Lecture 15), whose mean is exactly $\frac{k+1}{n+2}$.

22.5 Recipe

1. Write the likelihood $L(\theta) = \prod_i f(x_i; \theta)$, being careful about the range of validity.
2. Take logs: $\ell(\theta) = \sum_i \log f(x_i; \theta)$.
3. Differentiate, set to zero, solve.
4. Check it is a maximum (second derivative, or the shape of ℓ).
5. If the support depends on θ , skip steps 2–4 and argue directly.